

GENERAL CHEMISTRY

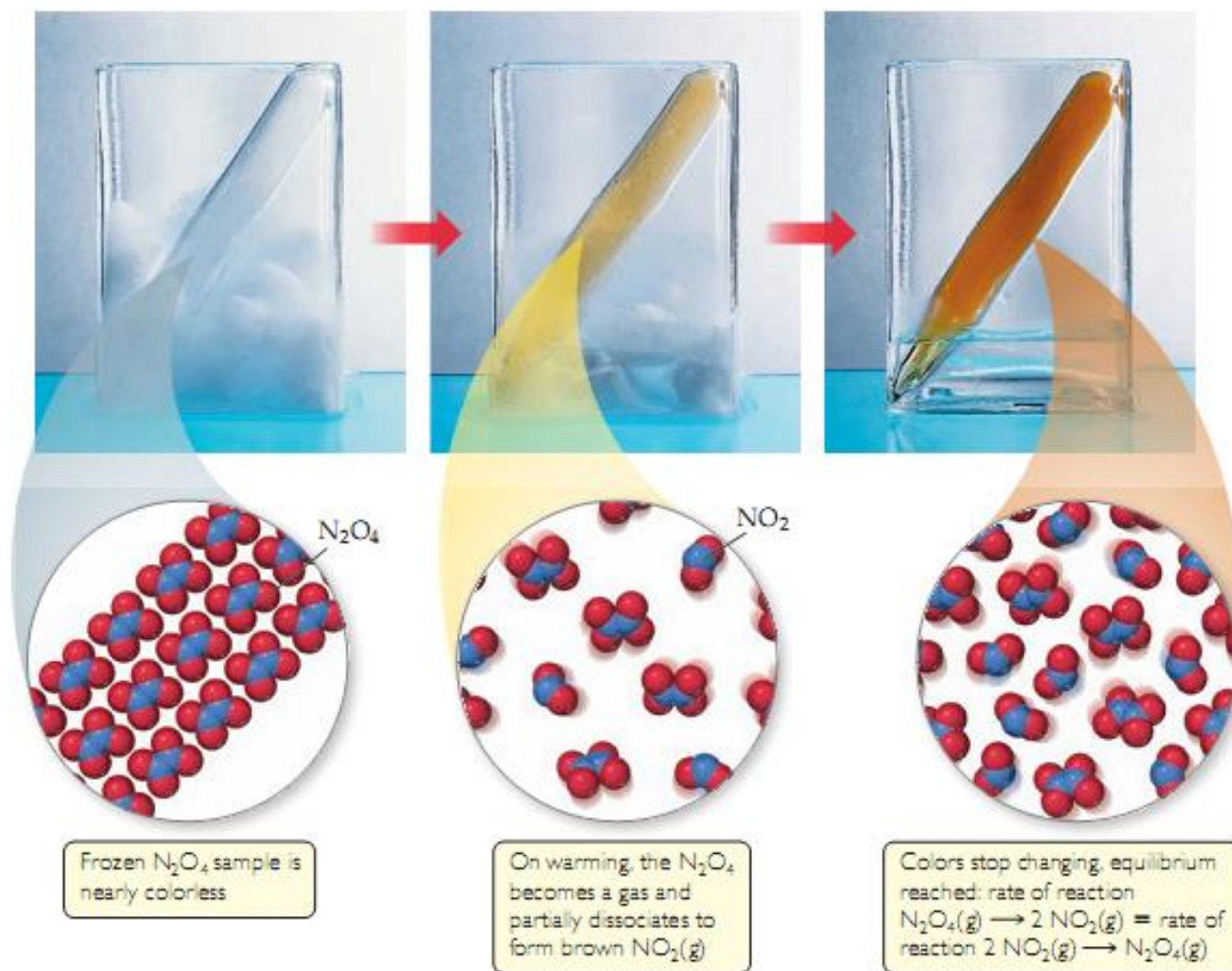
Chapter 15 Chemical Equilibrium

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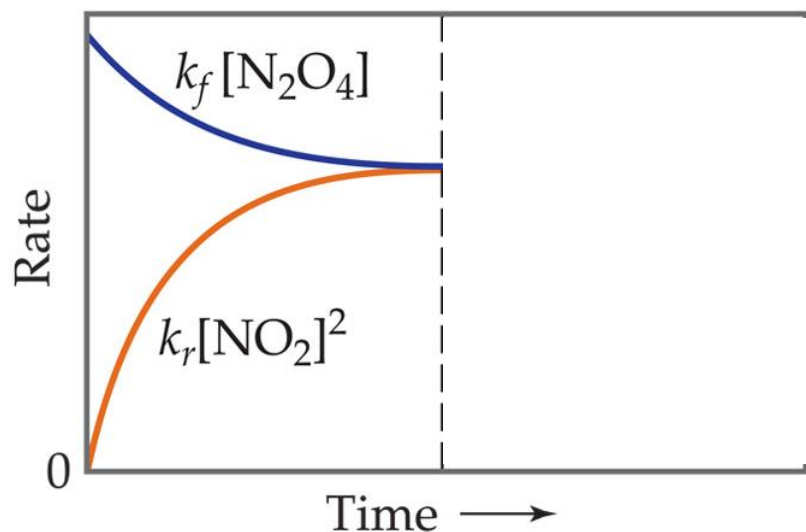
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15-1 The Concept of Equilibrium

Chemical equilibrium occurs when a reaction and its reverse reaction proceed at the same rate.



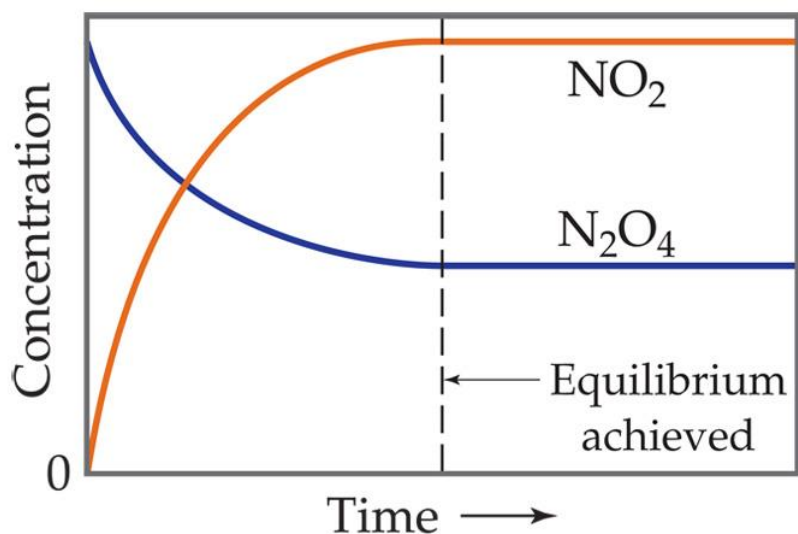
The Concept of Equilibrium



- ◆ As a system approaches equilibrium, both the forward and reverse reactions are occurring.



- ◆ At equilibrium, the forward and reverse reactions are proceeding *at the same rate*.



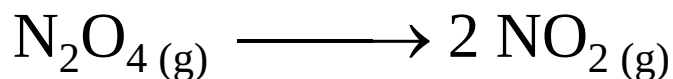
- ◆ Once equilibrium is achieved, the amount of each reactant and product remains constant.

Dynamic Equilibrium

- ◆ A dynamic equilibrium exists when the rates of the forward and reverse reactions are equal.
- ◆ There is no further change in [reactant] or [product].
- ◆ How fast you get to equilibrium depends on kinetics.

The Equilibrium Constant

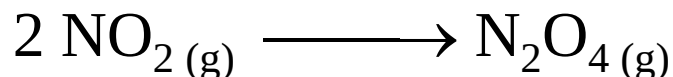
- ◆ Forward reaction:



Rate Law:

$$\text{Rate}_f = k_f [\text{N}_2\text{O}_4]$$

- ◆ Reverse reaction:



Rate Law:

$$\text{Rate}_r = k_r [\text{NO}_2]^2$$

- ◆ At equilibrium

$$\text{Rate}_f = \text{Rate}_r$$

$$k_f [\text{N}_2\text{O}_4] = k_r [\text{NO}_2]^2$$

$$\frac{k_f}{k_r} = \frac{[\text{NO}_2]^2}{[\text{N}_2\text{O}_4]}$$

The Equilibrium Constant

The ratio of the rate constants is a constant at that temperature, and the expression becomes

$$K_{\text{eq}} = \frac{k_f}{k_r} = \frac{[\text{NO}_2]^2}{[\text{N}_2\text{O}_4]}$$

Summary

1. At equilibrium, the concentrations of reactants and products no longer change with time.
2. For equilibrium to occur, neither reactants nor products can escape from the system.
3. At equilibrium a particular ratio of concentration terms equals a constant.

15-2 The Equilibrium Constant

- ◆ Consider the generalized reaction



- The equilibrium expression for this reaction would be

$$K_c = \frac{[C]^c [D]^d}{[A]^a [B]^b}$$

- Since pressure is proportional to concentration for gases in a closed system, the equilibrium expression can also be written

$$K_p = \frac{(P_C)^c (P_D)^d}{(P_A)^a (P_B)^b}$$

K_{eq} vs K_c vs K_p

- ◆ K_{eq} = the general expression for equilibrium constant expressions.
- ◆ $K_c = K_{eq}$ for which molar concentrations were used to evaluate the constant (i.e. subscript “c” = concentration).
- ◆ K_c includes
 - K_a (weak acids) and K_b (weak bases) (Chapter 16)
 - K_{sp} (solubility product) (Chapter 17)
 - K_w (water)
- ◆ $K_p = K_{eq}$ where “p” stands for pressure

Relationship Between K_c and K_p

- ◆ From the Ideal Gas Law we know that

$$PV = nRT$$

- Rearranging it, we get

$$P = \frac{nRT}{V}$$

If we express volume in liters the quantity (n/V) is equivalent to molarity.

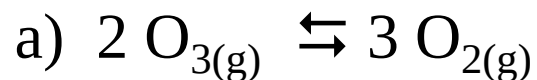
Plugging this into the expression for K_p for each substance, the relationship between K_c and K_p becomes

$$K_p = K_c (RT)^{\Delta n}$$

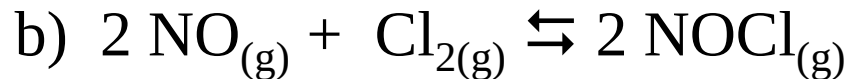
$\Delta n = (\text{moles of gaseous product}) - (\text{moles of gaseous reactant})$

EXAMPLE

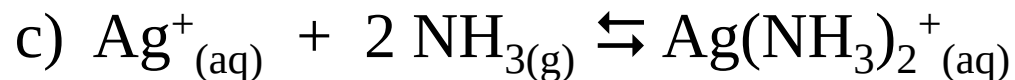
Write the equilibrium expression for K_{eq} for these three reactions:



$$K_c = \frac{[\text{O}_2]^3}{[\text{O}_3]^2}$$



$$K_c = \frac{[\text{NOCl}]^2}{[\text{NO}]^2[\text{Cl}_2]}$$



$$K_c = \frac{[\text{Ag}(\text{NH}_3)_2^+]}{[\text{Ag}^+][\text{NH}_3]^2}$$

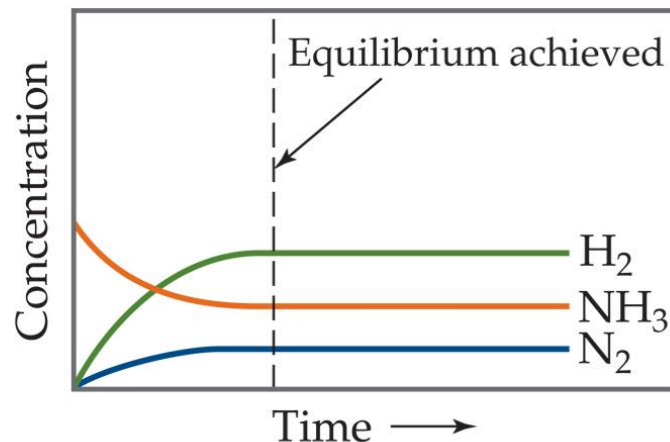
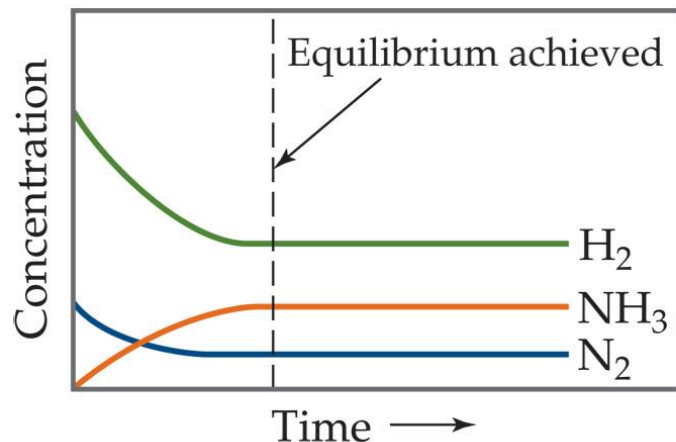
Equilibrium Can Be Reached from Either Direction

Table 15.1 Initial and Equilibrium Concentrations of $\text{N}_2\text{O}_4(g)$ and $\text{NO}_2(g)$ at $100\text{ }^\circ\text{C}$

Experiment	Initial $[\text{N}_2\text{O}_4]$ (M)	Initial $[\text{NO}_2]$ (M)	Equilibrium $[\text{N}_2\text{O}_4]$ (M)	Equilibrium $[\text{NO}_2]$ (M)	K_c
1	0.0	0.0200	0.00140	0.0172	0.211
2	0.0	0.0300	0.00280	0.0243	0.211
3	0.0	0.0400	0.00452	0.0310	0.213
4	0.0200	0.0	0.00452	0.0310	0.213

The ratio of $[\text{NO}_2]^2$ to $[\text{N}_2\text{O}_4]$ remains constant at this temperature no matter what the initial concentrations of NO_2 and N_2O_4 are.

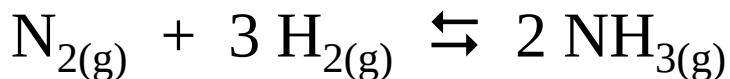
Equilibrium Can Be Reached from Either Direction



It doesn't matter whether we start with N_2 and H_2 or whether we start with NH_3 : we will have the same proportions of all three substances at equilibrium.

EXAMPLE

In the synthesis of ammonia from nitrogen and hydrogen,



$K_c = 9.60$ at 300°C , calculate K_p for this reaction at this temperature.

$$K_p = K_c (RT)^{\Delta n}$$

$$R = 0.08206 \text{ L.atm/mol.K}$$

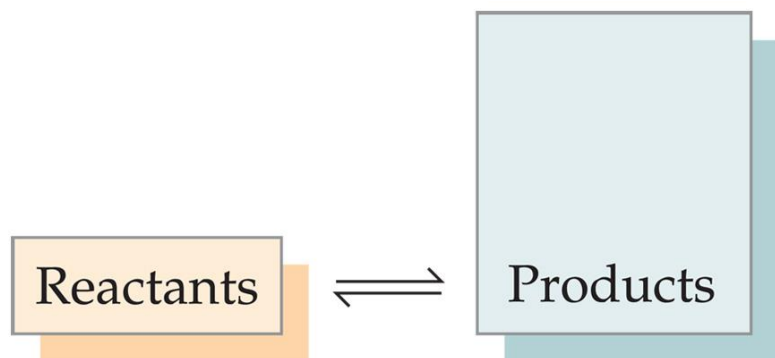
$$T = 300 + 273 = 573 \text{ K}$$

2 mol of gaseous products (2 NH_3) and 4 mol of gaseous reactants (1 N_2 + 3 H_2)

$$\Delta n = 2 - 4 = -2$$

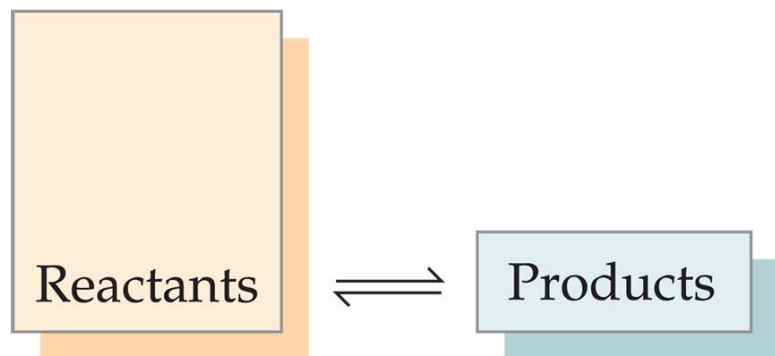
$$K_p = K_c (RT)^{\Delta n} = (9.60)(0.08206 \times 573)^{-2} = 4.34 \times 10^{-3}$$

15-3 Understanding and Working with Equilibrium Constants



$$K \gg 1$$

- ◆ If $K \gg 1$, the reaction is *product-favored*; product predominates at equilibrium.



$$K \ll 1$$

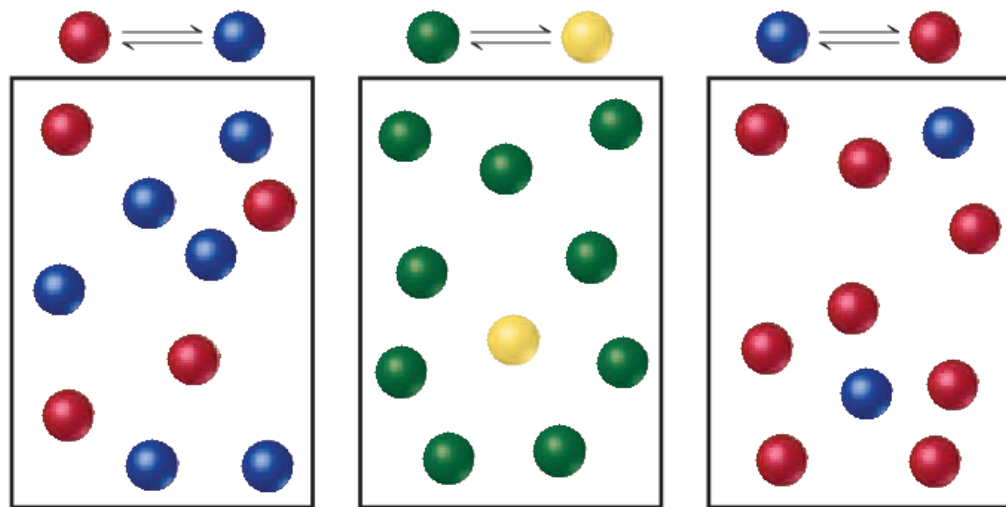
- If $K \ll 1$, the reaction is *reactant-favored*; reactant predominates at equilibrium.

EXAMPLE

The following diagrams represent three different systems at equilibrium, all in the same size containers.

a) Without doing any calculations, rank the three systems in order of increasing equilibrium constant, K_c .

b) If the volume of the containers is 1.0 L and each sphere represents 0.10 mol, calculate K_c for each system.



(i)

(ii)

(iii)

a) **(ii) < (i) < (iii)**

b) $K_c = \frac{[C]^c[D]^d}{[A]^a[B]^b}$

(i) $K_c = 0.60/0.40 = 1.5$

(ii) $K_c = 0.10/0.90$

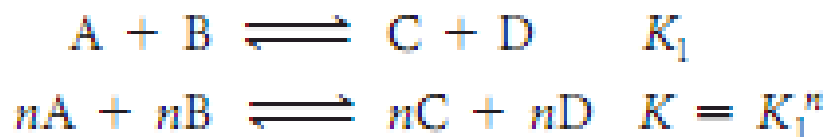
(iii) $K_c = 0.80/0.20 = 4.0$

Relating Chemical Equation Stoichiometry and Equilibrium Constants

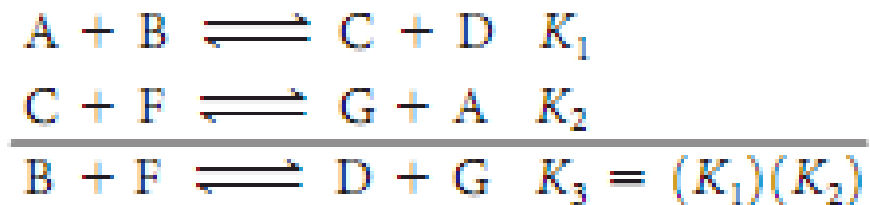
1. The equilibrium constant of a reaction in the **reverse** reaction is the **reciprocal** of the equilibrium constant of the **forward** reaction.



2. The equilibrium constant of a reaction that has been **multiplied** by a number is the equilibrium constant raised to a **power** that is equal to that number.



3. The equilibrium constant for a net reaction made up of **two or more steps** is the **product** of the equilibrium constants for the individual steps.



Manipulating Equilibrium Constants

1. The equilibrium constant of a reaction in the reverse reaction is the reciprocal of the equilibrium constant of the forward reaction.

In the opposite direction:



$$K_{\text{eq}} = \frac{[\text{NO}_2]^2}{[\text{N}_2\text{O}_4]} = 0.212$$

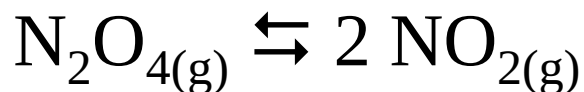


$$K_{\text{eq}} = \frac{[\text{N}_2\text{O}_4]}{[\text{NO}_2]^2} = 4.72$$

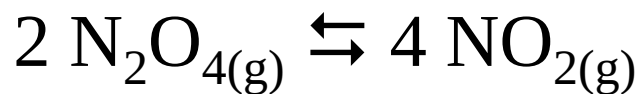
K_{eq} for these reactions are reciprocals of each other.

Manipulating Equilibrium Constants

2. The equilibrium constant of a reaction that has been multiplied by a number is the equilibrium constant raised to a power that is equal to that number.



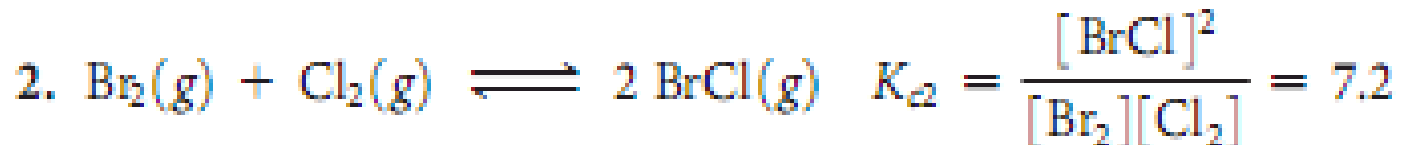
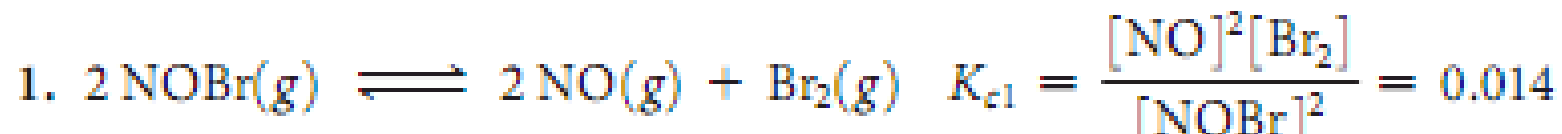
$$K_c = \frac{[\text{NO}_2]^2}{[\text{N}_2\text{O}_4]} = 0.212$$



$$K_c = \frac{[\text{NO}_2]^4}{[\text{N}_2\text{O}_4]^2} = (0.212)^2$$

Manipulating Equilibrium Constants

3. The equilibrium constant for a net reaction made up of two or more steps is the product of the equilibrium constants for the individual steps.



The net sum of these two equations is

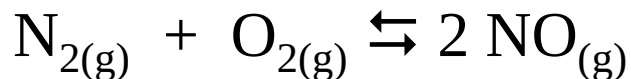


$$K_{c3} = \frac{[\text{NO}]^2[\text{BrCl}]^2}{[\text{NOBr}]^2[\text{Cl}_2]} = \frac{[\text{NO}]^2[\text{Br}_2]}{[\text{NOBr}]^2} \times \frac{[\text{BrCl}]^2}{[\text{Br}_2][\text{Cl}_2]}$$

$$K_{c3} = (K_{c1})(K_{c2}) = (0.014)(7.2) = 0.10$$

EXAMPLE

The equilibrium constant for the reaction of N_2 with O_2 to form NO equals $K_c = 1 \times 10^{-30}$ at 25°C .



Using this information, write the equilibrium constant expression and calculate the equilibrium constant for the following reaction:



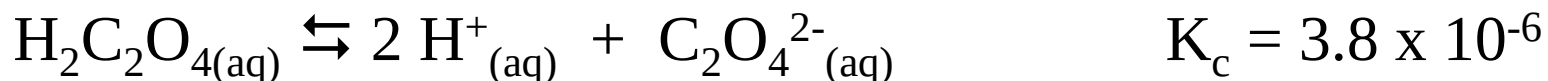
$$K_{\text{c forward}} = \frac{[\text{NO}]^2}{[\text{N}_2][\text{O}_2]}$$

$$K_{\text{c reverse}} = \frac{[\text{N}_2][\text{O}_2]}{[\text{NO}]^2}$$

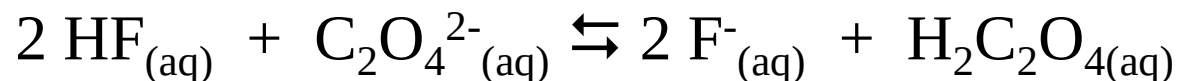
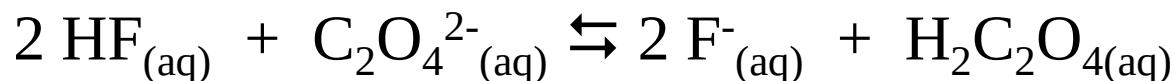
$$K_{\text{c reversed}} = 1/K_{\text{c forward}} = 1/(1 \times 10^{-30}) = 10^{30}$$

EXAMPLE

Given the following information,



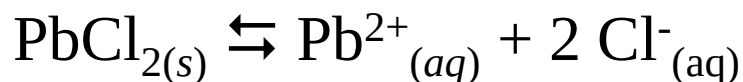
determine the value of K_c for the following reaction:



$$K_c = (4.6 \times 10^{-7})(2.6 \times 10^5) = 0.12$$

15-4 Heterogeneous Equilibria

- ◆ Equilibria in which all reactants and products are present in the same phase are called **homogeneous equilibria**.
- ◆ Equilibria in which one or more reactants or products are present in a different phase are called **heterogeneous equilibria**.

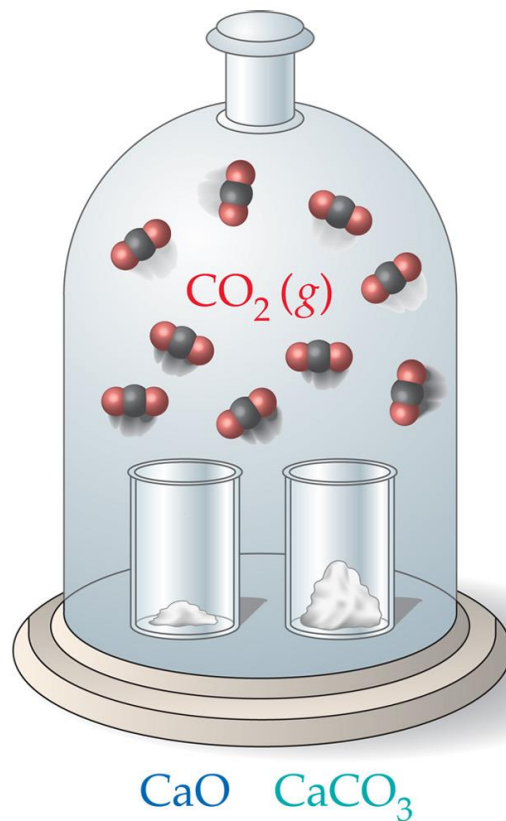
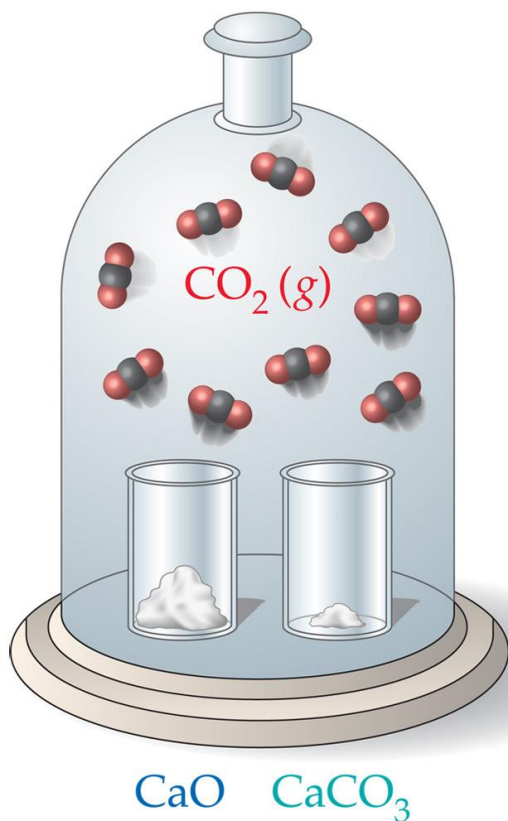


$$K_c = [\text{Pb}^{2+}] [\text{Cl}^{-}]^2$$

Whenever a pure solid or a pure liquid is involved in a heterogeneous equilibrium, its concentration is not included in the equilibrium constant expression

Heterogeneous Equilibria

Pure solids & gases



At a given temperature, the equilibrium pressure of CO₂ in the bell jars is the same no matter how much of each solid is present

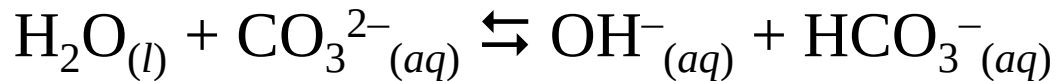
$$K_c = [\text{CO}_2] \text{ and } K_p = P_{\text{CO}_2}$$

Heterogeneous Equilibria

Low solute concentration

Systems where the solvent is involved as a reactant or product and the solutes are present at **low** concentrations.

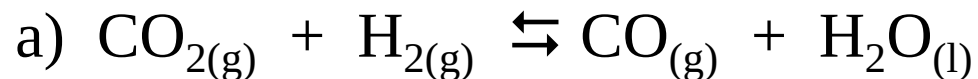
e.g. dissociation of a weak acid



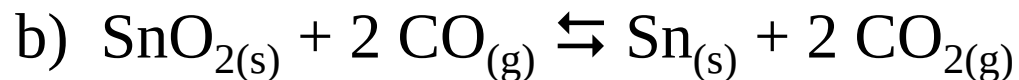
$$K_c = [\text{OH}^{-}][\text{HCO}_3^{-}] / [\text{CO}_3^{2-}]$$

EXAMPLE

Write the equilibrium-constant K_c for each of the following reactions:



$$K_c = \frac{[\text{CO}]}{[\text{CO}_2][\text{H}_2]}$$



$$K_c = \frac{[\text{CO}_2]^2}{[\text{CO}]^2}$$

EXAMPLE

Each of these mixtures was placed in a closed container and allowed to stand:

- (a) $\text{CaCO}_3(\text{s})$
- (b) $\text{CaO}(\text{s})$ and $\text{CO}_2(\text{g})$ at a pressure greater than the value of K_p
- (c) $\text{CaCO}_3(\text{s})$ and $\text{CO}_2(\text{g})$ at a pressure greater than the value of K_p
- (d) $\text{CaCO}_3(\text{s})$ and $\text{CaO}(\text{s})$

Determine whether or not each mixture can attain the equilibrium



$$K_p = P_{\text{CO}_2}$$

- (a) CaCO_3 simply decomposes, forming $\text{CaO}(\text{s})$ and $\text{CO}_2(\text{g})$ until the equilibrium pressure of CO_2 is attained. There must be enough CaCO_3 , however, to allow the CO_2 pressure to reach equilibrium.
- (b) CO_2 continues to combine with CaO until the partial pressure of the CO_2 decreases to the equilibrium value.
- (c) Because there is no CaO present, equilibrium cannot be attained; there is no way the CO_2 pressure can decrease to its equilibrium value (which would require some CO_2 to react with CaO).
- (d) The situation is essentially the same as in (a)

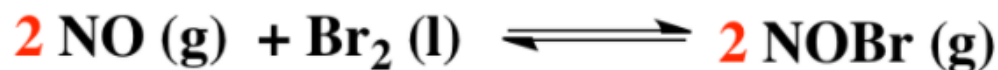
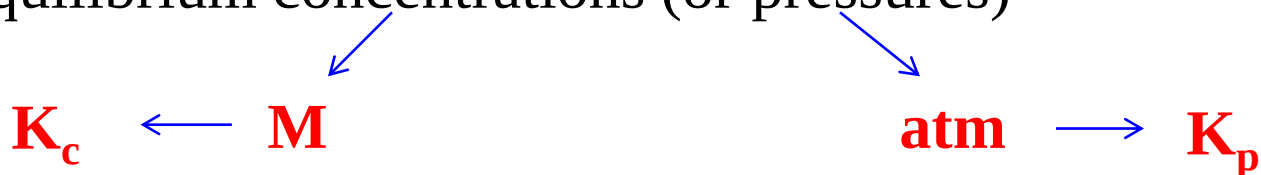
15-5 Calculating Equilibrium Constants Using ICE charts

An ICE Chart should be used when we are **missing more than one** equilibrium amount for compounds in our balanced equation.

I = initial concentrations (or pressures) for each species

C = change in the concentrations (or pressures) for each species

E = equilibrium concentrations (or pressures)



Initial	0.10 M or atm		0
Change	- 2 x		+ 2 x
Equilibrium	0.10 - 2 x		+ 2 x

Equilibrium Calculations

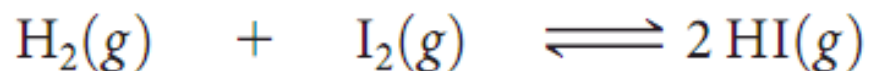
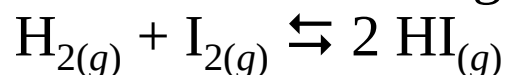
Often, we do not know the equilibrium concentrations of all species in an equilibrium mixture. If we know the initial concentrations and the equilibrium concentration of at least one species, then:

1. ***Tabulate all known initial and equilibrium concentrations*** of the species that appear in the equilibrium-constant expression.
2. For those species for which initial and equilibrium concentrations are known, ***calculate the change in concentration*** that occurs as the system reaches equilibrium.
3. Use the stoichiometry of the reaction (that is, the coefficients in the balanced chemical equation) to ***calculate the changes in concentration for all other species*** in the equilibrium-constant expression.
4. Use initial concentrations from step 1 and changes in concentration from step 3 to ***calculate any equilibrium concentrations not tabulated*** in step 1.
5. ***Determine the value of the equilibrium constant.***

EXAMPLE

A closed system initially containing $1.000 \times 10^{-3} \text{ M H}_2$ and $2.000 \times 10^{-3} \text{ M I}_2$ at 448°C is allowed to reach equilibrium. Analysis of the equilibrium mixture shows that the concentration of HI is $1.87 \times 10^{-3} \text{ M}$.

Calculate K_c at 448°C for the reaction taking place, which is



Initial concentration (M)	1.000×10^{-3}	2.000×10^{-3}	0
Change in concentration (M)	-x	-x	+2x
Equilibrium concentration (M)	$1.000 \times 10^{-3} - x$	$2.000 \times 10^{-3} - x$	1.87×10^{-3}

$$+2x = 1.87 \times 10^{-3}$$

$$x = 1.87 \times 10^{-3} / 2 = 9.35 \times 10^{-4}$$

$$[\text{H}_2] = 1.000 \times 10^{-3} \text{ M} - 9.35 \times 10^{-4} \text{ M} = 0.065 \times 10^{-3} \text{ M}$$

$$[\text{I}_2] = 2.000 \times 10^{-3} \text{ M} - 9.35 \times 10^{-4} \text{ M} = 1.065 \times 10^{-3} \text{ M}$$

$$K_c = \frac{[\text{HI}]^2}{[\text{H}_2][\text{I}_2]} = \frac{(1.87 \times 10^{-3})^2}{(0.065 \times 10^{-3})(1.065 \times 10^{-3})} = 51$$

15-6 Applications of Equilibrium

- ◆ Predicting the Direction of Reaction
- ◆ Calculating Equilibrium Concentrations

The Reaction Quotient (Q)

- ◆ Q gives the same ratio the equilibrium expression gives, but for a system that is *not* at equilibrium.
- ◆ To calculate Q, one substitutes the initial concentrations of reactants and products into the equilibrium expression.

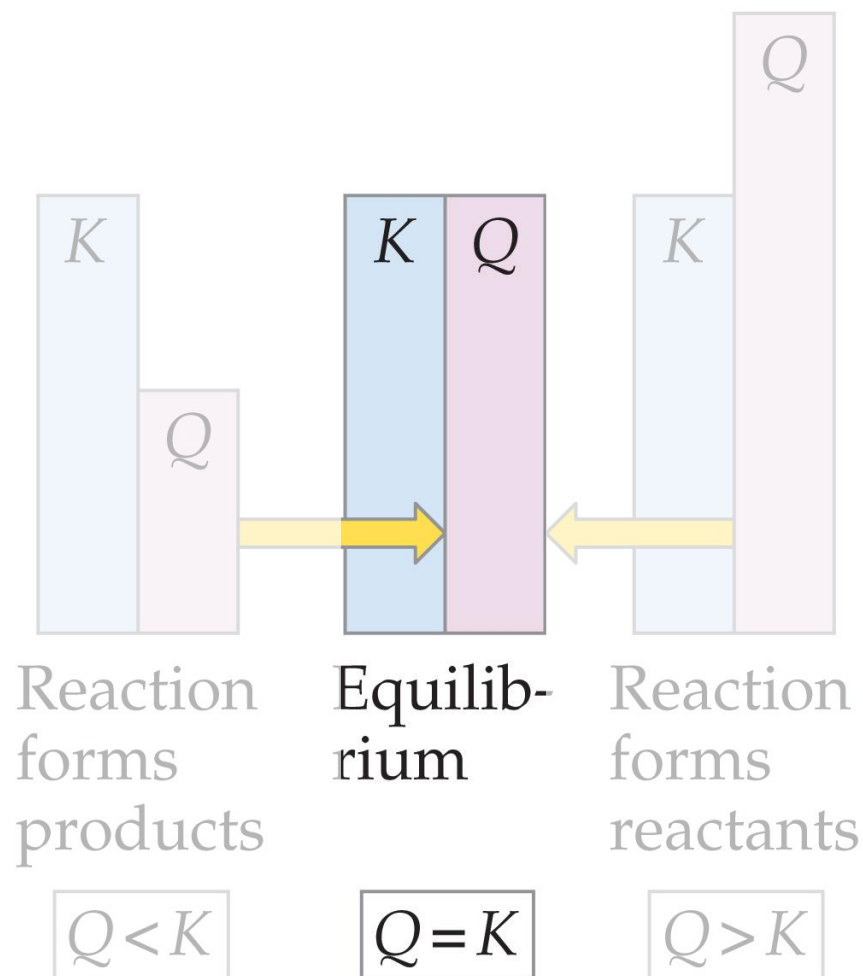
$$Q = \frac{[C]^c [D]^d}{[A]^a [B]^b}$$

The Reaction Quotient

- ◆ Note: $Q = K_{\text{eq}}$ only at equilibrium.
- ◆ If $Q < K_{\text{eq}}$ then the forward reaction must occur to reach equilibrium.
- ◆ If $Q > K_{\text{eq}}$ then the reverse reaction must occur to reach equilibrium.
Products are consumed, reactants are formed.
 Q decreases until it equals K_{eq} .

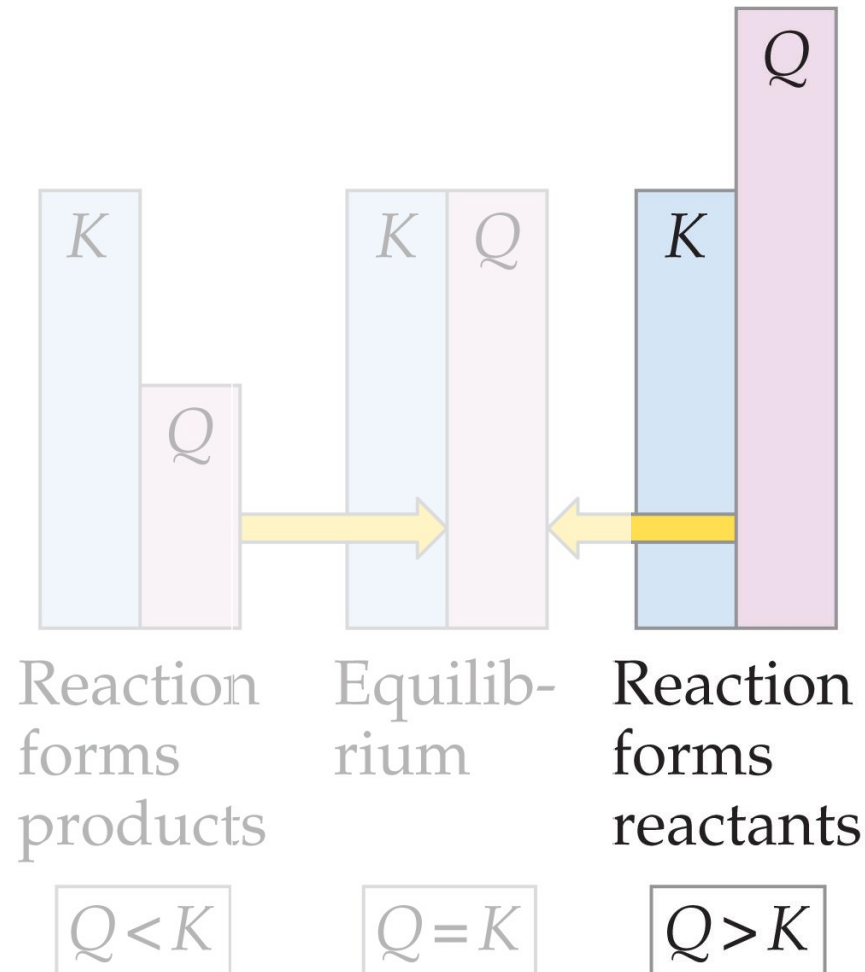
If $Q = K$,

The system is at equilibrium.



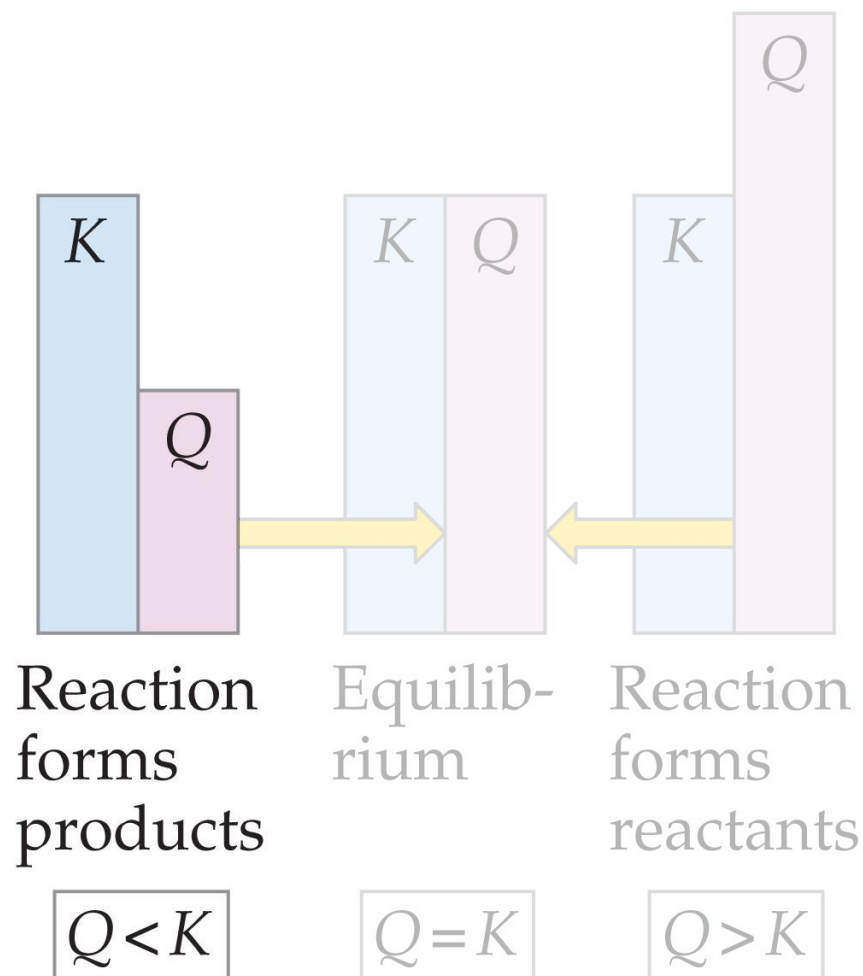
If $Q > K$,

There is too much product, and the equilibrium shifts to the left.



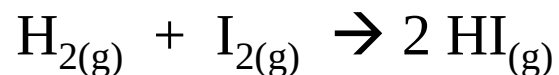
If $Q < K$,

There is too much reactant, and the equilibrium shifts to the right.



EXAMPLE

At 448°C the equilibrium constant, K_c , for the reaction



is 50.5.

Predict how the reaction will proceed to reach equilibrium at 448°C if we start with 2.0×10^{-2} mol of HI, 1.0×10^{-2} mole of H_2 , and 3.0×10^{-2} mol of I_2 in a 2.00-L container.

The initial concentrations are

$$[\text{HI}] = 2.0 \times 10^{-2} \text{ mol} / 2.00 \text{ L} = 1.0 \times 10^{-2} \text{ M}$$

$$[\text{H}_2] = 1.0 \times 10^{-2} \text{ mol} / 2.00 \text{ L} = 5.0 \times 10^{-3} \text{ M}$$

$$[\text{I}_2] = 3.0 \times 10^{-2} \text{ mol} / 2.00 \text{ L} = 1.5 \times 10^{-2} \text{ M}$$

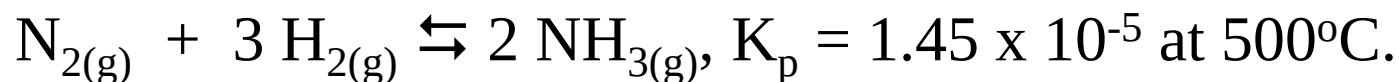
The reaction quotient is therefore

$$Q_c = \frac{[\text{HI}]^2}{[\text{H}_2][\text{I}_2]} = \frac{(1.0 \times 10^{-2})^2}{(5.0 \times 10^{-3})(1.5 \times 10^{-2})} = 1.3$$

Because $Q_c < K_c$, the reaction proceeds left to right to attain equilibrium

EXAMPLE

For the Haber process,



In an equilibrium mixture of the three gases at 500°C , the partial pressure of H_2 is 0.928 atm and that of N_2 is 0.432 atm. What is the partial pressure of NH_3 in this equilibrium mixture?



Equilibrium pressure (atm)	0.432	0.928	x
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$$K_p = \frac{(P_{\text{NH}_3})^2}{P_{\text{N}_2}(P_{\text{H}_2})^3} = \frac{x^2}{(0.432)(0.928)^3} = 1.45 \times 10^{-5}$$

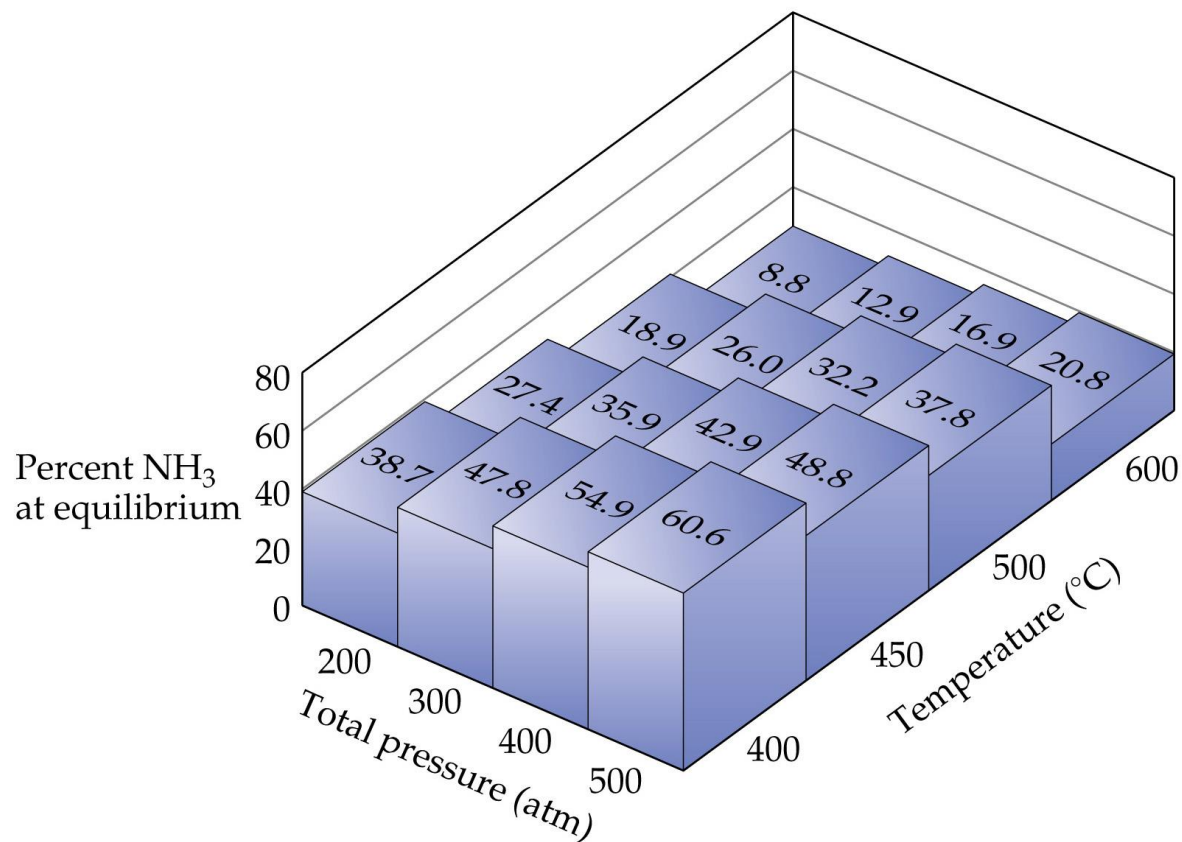
$$x^2 = (1.45 \times 10^{-5})(0.432)(0.928)^3 = 5.01 \times 10^{-6}$$

$$x = \sqrt{5.01 \times 10^{-6}} = 2.24 \times 10^{-3} \text{ atm} = P_{\text{NH}_3}$$

15-7 Le Châtelier's Principle

“If a system at equilibrium is disturbed by a change in temperature, pressure, or the concentration of one of the components, the system will **shift its equilibrium position so as to counteract the effect of the disturbance.**”

The Haber Process



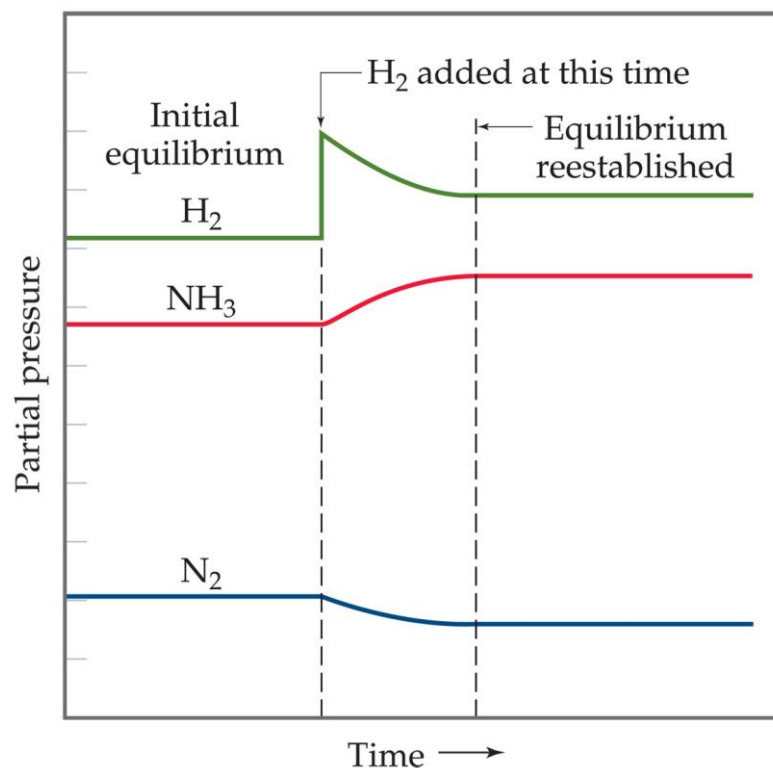
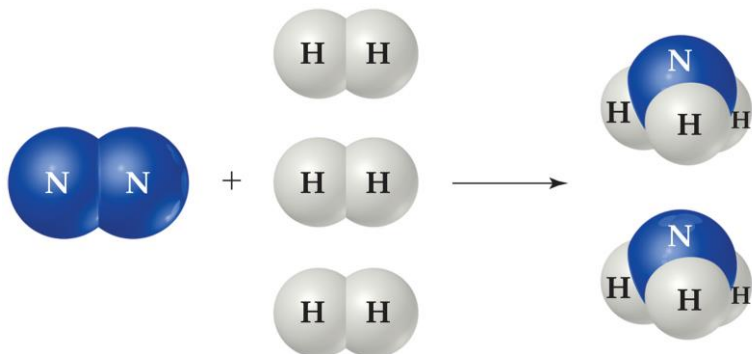
As P increases, the amount of ammonia present at equilibrium increases.

As T increases, the amount of ammonia at equilibrium decreases.

Can this be predicted?

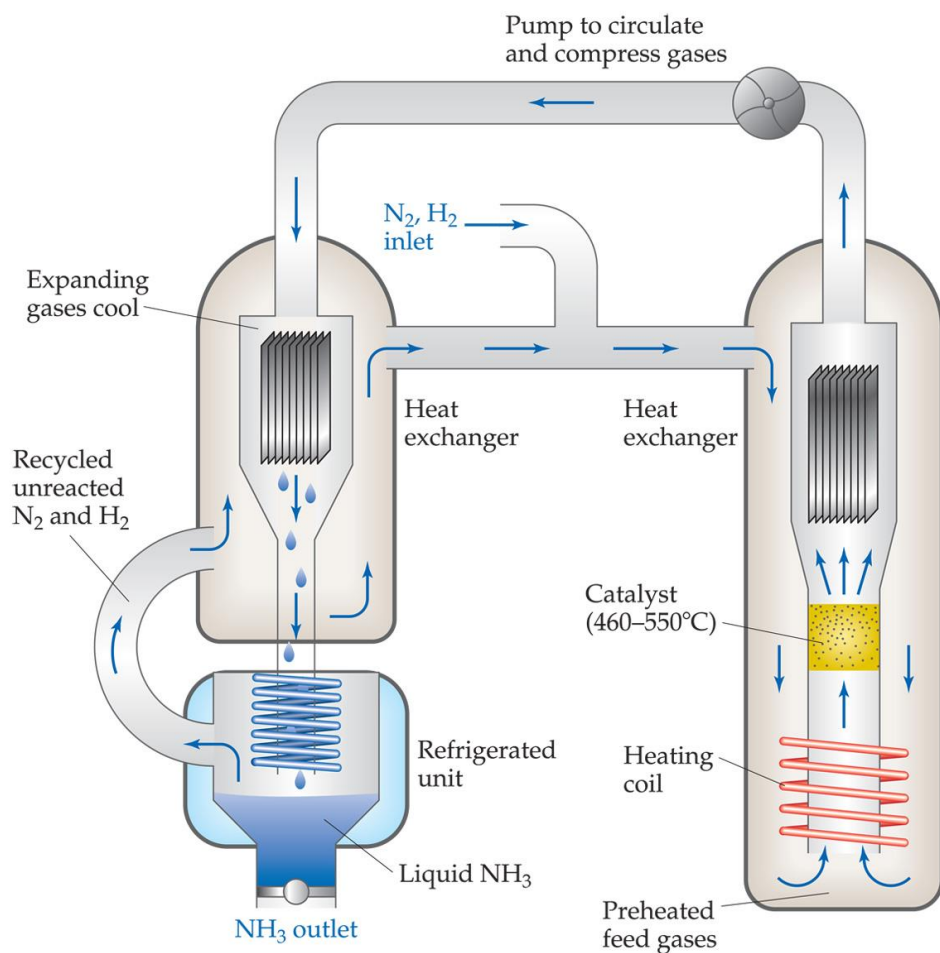
The Haber Process

The transformation of nitrogen and hydrogen into ammonia (NH_3) is of tremendous significance in agriculture, where ammonia-based fertilizers are of utmost importance.



If H_2 is added to the system, N_2 will be consumed and the two reagents will form more NH_3 .

The Haber Process



This apparatus helps push the equilibrium to the right by removing the ammonia (NH₃) from the system as a liquid.

Effects of Volume and Pressure Changes

Le Châtelier's principle predicts that if pressure is increased, the system will shift to counteract the increase.

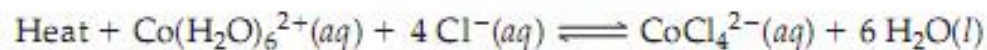
- ◆ That is, the system shifts to remove gases and decrease pressure.
- ◆ An increase in pressure favors the direction that has fewer moles of gas.

Effects of Volume and Pressure Changes

- ◆ In a reaction with the same number of moles of gas in the products and reactants, changing the pressure has no effect on the equilibrium.
- ◆ No change will occur if we increase the total gas pressure by the addition of a gas that is not involved in the reaction.

The Effect of Changes in Temperature

$\Delta H > 0$, endothermic reaction



Pink

Blue

The diagram illustrates the effect of temperature changes on the equilibrium between two cobalt complexes. It features three beakers connected by arrows indicating the direction of temperature change. Above each beaker is a circular inset showing the molecular models of the species in equilibrium.

- Left Beaker (Pink):** The solution is pink. The inset shows a high concentration of pink $\text{Co}(\text{H}_2\text{O})_6^{2+}$ ions and a low concentration of blue CoCl_4^{2-} ions. A pink arrow points from the central beaker to this one, labeled "Cool".
- Center Beaker (Violet):** The solution is violet. The inset shows a mixture of pink $\text{Co}(\text{H}_2\text{O})_6^{2+}$ and blue CoCl_4^{2-} ions. A blue arrow labeled "Cool" points from this beaker to the left one, and a red arrow labeled "Heat" points from this beaker to the right one.
- Right Beaker (Blue):** The solution is blue. The inset shows a high concentration of blue CoCl_4^{2-} ions and a low concentration of pink $\text{Co}(\text{H}_2\text{O})_6^{2+}$ ions. A blue arrow points from the central beaker to this one, labeled "Heat".

Explanatory Text:

- Left Beaker:** Solution appears pink because lowering the temperature shifts the equilibrium to favor formation of the pink $\text{Co}(\text{H}_2\text{O})_6^{2+}$ ion.
- Center Beaker:** Solution appears violet because appreciable amounts of both pink $\text{Co}(\text{H}_2\text{O})_6^{2+}$ and blue CoCl_4^{2-} are present.
- Right Beaker:** Solution appears blue because raising the temperature shifts the equilibrium to favor formation of the blue CoCl_4^{2-} ion.

The Effect of Changes in Temperature

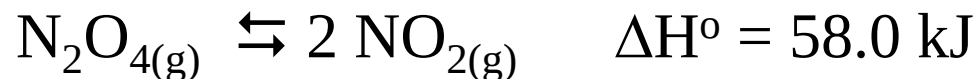
Endothermic: Reactants + heat $\rightarrow$ products

Exothermic: Reactants $\rightarrow$ products + heat

When the temperature of a system at equilibrium is increased, the system reacts as if we added a reactant to an endothermic reaction or a product to an exothermic reaction. The equilibrium shifts in the direction that consumes the excess reactant (or product), namely heat.

EXAMPLE

Consider the following equilibrium:

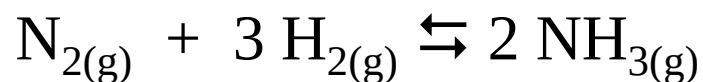


In what direction will the equilibrium shift when each of the following changes is made to a system at equilibrium:

- a) add N_2O_4 $\rightarrow$
- b) remove NO_2 $\rightarrow$
- c) increase the total pressure by adding $\text{N}_{2(g)}$ **No shift**
- d) increase the volume $\rightarrow$
- e) decrease the temperature? $\leftarrow$

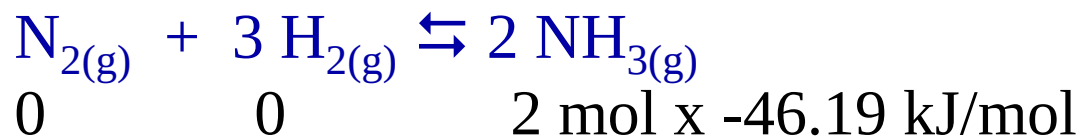
EXAMPLE

a) Using the standard heat of formation, determine the standard enthalpy change for the reaction



b) Determine how the equilibrium constant for this reaction should change with temperature.

a)



$$\Delta H_{\text{rctn}} = -92.38 \text{ kJ/mol}$$

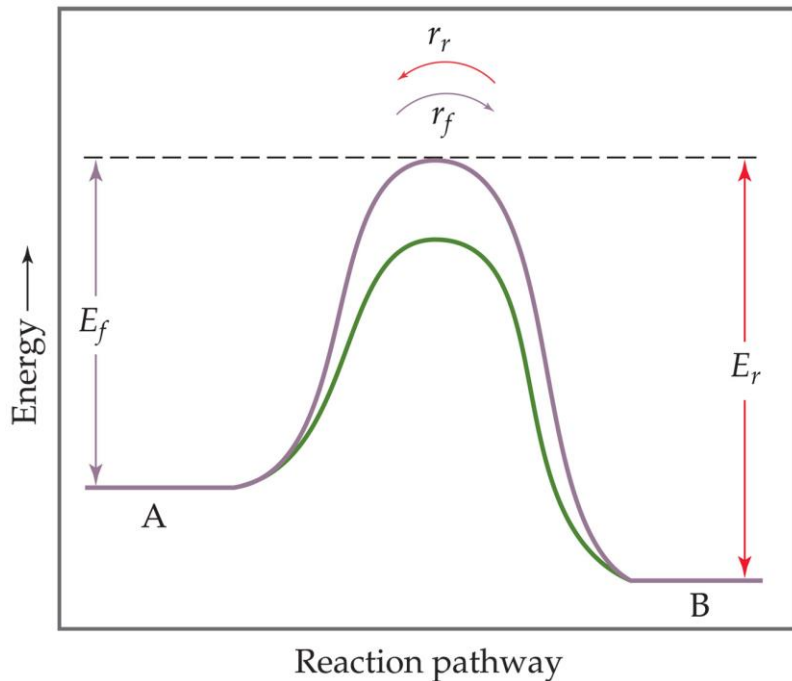
Substance	ΔH_f° (kJ/mol)
H(g)	217.94
H ⁺ (aq)	0
H ⁺ (g)	1536.2
H ₂ (g)	0
N(g)	472.7
N ₂ (g)	0
NH ₃ (aq)	-80.29
NH ₃ (g)	-46.19

b) Because the reaction in the forward direction is exothermic, we can consider heat a product of the reaction. An increase in temperature causes the reaction to shift in the direction of less NH₃ and more N₂ and H₂.

TABLE 15.2 ■ Variation in K_p for the Equilibrium $\text{N}_2 + 3 \text{H}_2 \rightleftharpoons 2 \text{NH}_3$ as a Function of Temperature

Temperature (°C)	K_p
300	4.34×10^{-3}
400	1.64×10^{-4}
450	4.51×10^{-5}
500	1.45×10^{-5}
550	5.38×10^{-6}
600	2.25×10^{-6}

The Effect of Catalysts



- ◆ Catalysts increase the rate of both the forward *and* reverse reactions.
- ◆ When one uses a catalyst, equilibrium is achieved faster, but the equilibrium composition remains unaltered.

Homeworks

15.93

15.94

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